

Frame Numbers	FAS/NFAS Bits										Ch1	Ch15	Ch16										Ch17	Ch19
											Ch1	Ch15	MFAS Bits										Ch16	Ch19
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
0	C1	0	0	1	1	0	1	1	1	1			0	0	0	0	S	A	S	S				
1	0	1	A	S	S	S	S	S	S				a1	b1	c1	d1	a16	b16	c16	d16				
2	C2	0	0	1	1	0	1	1	1	1			a2	b2	c2	d2	a17	b17	c17	d17				
3	0	1	A	S	S	S	S	S	S				a3	b3	c3	d3	a18	b18	c18	d18				
4	C3	0	0	1	1	0	1	1	1	1			a4	b4	c4	d4	a19	b19	c19	d19				
5	1	1	A	S	S	S	S	S	S				a5	b5	c5	d5	a20	b20	c20	d20				
6	C4	0	0	1	1	0	1	1	1	1			a6	b6	c6	d6	a21	b21	c21	d21				
7	0	1	A	S	S	S	S	S	S				a7	b7	c7	d7	a22	b22	c22	d22				
8	C1	0	0	1	1	0	1	1	1	1			a8	b8	c8	d8	a23	b23	c23	d23				
9	1	1	A	S	S	S	S	S	S				a9	b9	c9	d9	a24	b24	c24	d24				
10	C2	0	0	1	1	0	1	1	1	1			a10	b10	c10	d10	a25	b25	c25	d25				
11	1	1	A	S	S	S	S	S	S				a11	b11	c11	d11	a26	b26	c26	d26				
12	C3	0	0	1	1	0	1	1	1	1			a12	b12	c12	d12	a27	b27	c27	d27				
13	E	1	A	S	S	S	S	S	S				a13	b13	c13	d13	a28	b28	c28	d28				
14	C4	0	0	1	1	0	1	1	1	1			a14	b14	c14	d14	a29	b29	c29	d29				
15	E	1	A	S	S	S	S	S	S				a15	b15	c15	d15	a30	b30	c30	d30				
Timeslot TS0											TS1	TS15											TS17	TS31
Legend																								
C1											Cyclic redundancy checksum bits													
E											CRC-4 Error signaling bits													
											CRC-4 Multiframe alignment													
											Frame alignment bits													
											Frame alignment supervision bit													
											CAS multiframe alignment													

Fig. 3: The 2.048Mbps basic frame structure

resulting in $32 \times 64\text{ kbps} = 2.048\text{ Mbps}$ speed.

In CCS scheme, TS0 and TS16 time slots are used for synchronisation, bit-error detection, alarm indication, frame alignment, etc. Cyclic redundancy check (CRC-4) bits allow the detection of errors. In TS0 time slot, frame alignment supervision (FAS) bits allow targeting of synchronisation to find the beginning of the frame. FAS bits are only transmitted on odd frames. Non-frame alignment supervision (NFAS) is used to manage alarms and errors, such as loss of signal (LOS) indication in the event of link failure or frame loss. NFAS uses a bit equal to '1' to avoid coincidences. In TS16, bit A is used for remote alarm indication for such instances as a power fault, loss of incoming signal, or loss of multi frame alignment. Bit S is used for maintenance or performance monitoring. Multi frame alignment signal (MFAS) is used to synchronise the channel-associated signalling.

Difficulties with PDH

Traditionally, PDH system is pleio-synchronous, which means the network elements work more or less in

time-synchronised manner, not in exact sync. Slight differences in timing signals mean that justification or stuffing is necessary when forming the multiplexed signals. Inserting or dropping an individual 64kbit/s channel to or from a higher digital hierarchy requires a considerable amount of complex multiplexer/de-multiplexer equipment.

One of the major hurdles with PDH system is adoption of different standards around the world. Various geographies of the world use different hierarchies, which lead to problems of international interworking. For example, between those countries using 1.544Mbit/s systems (USA and Japan) and those using the 2.048Mbit/s system (Europe), specialised interface equipment is required to interwork the two hierarchies.

To recover a 64kbit/s channel from a 140Mbit/s PDH signal, it is necessary to de-multiplex the signal all the way down to the 2Mbit/s level before the location of the 64kbit/s channel can be identified. PDH requires 'steps' (140-34, 34-8, 8-2 de-multiplex; 2-8, 8-34, 34-140 multiplex) to drop out or add an individual speech or data channel (Fig. 1).

Channel cross-connection is also an issue with PDH. In this system, identification of individual channels in a higher-order bit stream is not possible. Most PDH network equipment is proprietary. Vendors use their own line coding, optical interfaces, etc. Moreover, PDH systems lack in network management and monitoring capabilities too. In most parts of the world, there is no standardised definition of PDH bit rates greater than 140Mbit/s. Besides, each multiplexing section has to add overhead bits for justification, which is again not an efficient use of the available transmission bandwidth. PDH network architecture does not allow ring structure and works in point-to-point topology.

Conclusion

A PDH transport system provides the technical means to transfer large quantities of data between two network nodes. But it is very difficult to interoperate different PDH networks as it is a specific vendor proprietary based technology that lacks in global standardisation. Different parts of the world use different hierarchies for multiplexing and de-multiplexing. Moreover, PDH does not allow direct multiplexing, which means individual tributary signals cannot be inserted or removed into the PDH multiplexed signal without intermediate multiplexing and de-multiplexing steps. Supervision and maintenance functions are limited. This makes PDH an inefficient signal transport technology. ●

The author, working with BSNL, holds a PhD degree from IIT (BHU), Varanasi. He holds senior research fellowship of UGC at Centre of Advanced Study, Department of Electronics Engineering, IT-BHU in Varanasi. His current research interests include wired and wireless technologies for high-speed Internet access